HófvarpnirHCON: A Fast Dictionary-Based Crystal Density Predictor

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Abstract

We introduce HófvarpnirHCON, a fast, interpretable, dictionary-based crystal density predictor that achieves accuracy comparable to state-of-the-art machine learning models (pooled MAE ≈ 0.035 g cm^{-3} across more than 235,000 molecular structures) while requiring no GPU, no neural networks, and no quantum mechanical calculations. The method is built on a simple physical insight: the effective crystal volume can be approximated by a composition-derived reference volume, the Leonardus Volume $V_L = C_{\mathrm{LV}} \sum_i m_i^{1/3}$ with $C_{\mathrm{LV}} = 6.1$, minus empirically determined bond-specific packing-overlap corrections. By precomputing these corrections as lookup tables via non-negative least squares (NNLS) and stratifying by molecular weight regime, we achieve inference rates exceeding 1,600 predictions per second on a standard CPU core. The same framework handles co-crystals via mass-weighted averaging of component densities, with no additional parameters or model modifications. The code is open-source under the BSD-3 license.

1 Introduction

Crystal density is a fundamental descriptor in crystal engineering, influencing packing efficiency, polymorph stability, solubility, and mechanical behaviour. Accurate density prediction is therefore essential for pharmaceutical development, energetic materials research, and high-throughput screening of functional organic crystals. Despite its importance, reliable prediction remains challenging because it depends on both molecular structure and intermolecular packing.

Traditional approaches fall into three broad categories. First, empirical additivity methods and van der Waals volume-based estimates are computationally fast but often suffer from systematic errors because they do not account for bond-specific volume contractions or packing variations. Second, machine learning models—including random forests, gradient-boosted trees, and kernel methods—achieve higher accuracy but require large curated datasets and periodic retraining, and they often lack physical interpretability. Third, graph neural networks (GNNs) and deep learning models have shown impressive performance, yet they rely on tensor operations, often require GPU acceleration, and are sensitive to training data biases.

At the highest level of accuracy, density functional theory (DFT) and periodic crystal-structure prediction (CSP) workflows can predict crystal densities from first principles. However, these methods are computationally expensive, often requiring hours to days per molecule, and are impractical for high-throughput screening of thousands or millions of candidates.

Here we introduce HófvarpnirHCON, a fast, interpretable, dictionary-based crystal density predictor. The bond-overlap dictionaries are trained once from experimental data using Non-Negative Least Squares (NNLS), a lightweight linear optimization that converges rapidly and requires no GPU, no neural networks, and no quantum mechanical calculations. After training, predictions reduce to descriptor generation and dictionary evaluation, enabling inference at speeds exceeding 1,600 predictions per second on standard commodity hardware.

The framework was evaluated across six published crystal-density datasets containing experimentally measured crystal structures and comprising more than 235,000 molecular structures, achieving a pooled molecule-weighted mean absolute error of 0.035 g cm^{-3}.

In this document, we describe the model, its implementation, and its validation on publicly available experimental datasets. The code is open-source under the BSD-3 license.

2 Methods

2.1 Overview of the Prediction Framework

HófvarpnirHCON predicts crystal density from a SMILES string using a two-step procedure. First, the Leonardus Volume is calculated from the sum of cube roots of atomic masses, scaled by a fixed global factor. Second, bond-specific packing-overlap corrections—precomputed and stored in molecular-weight-stratified dictionaries—are subtracted from this reference volume to obtain the corrected crystal volume. Density is then computed as molecular weight divided by the corrected volume.

The central volume decomposition is:

$$V_{\text{crystal}} \approx V_L - \Delta V_{\text{bond}}$$

where V_L is the composition-derived Leonardus Volume and ΔV_{bond} represents the sum of bond-level correction terms.

For co-crystals (SMILES strings containing a dot character), the predictor recursively computes densities for each component and returns the mass-weighted average.

2.2 Leonardus Volume

The starting point is the Leonardus Volume, a composition-derived reference volume calculated directly from atomic masses:

$$V_L = C_{\text{LV}} \sum_i m_i^{1/3}$$

where:

- $C_{\text{LV}} = 6.1$ is the fixed global scaling factor,
- m_i is the atomic mass of atom i,
- the sum runs over all atoms in the molecule (including explicit hydrogens).

The cubic-root transformation $m_i^{1/3}$ provides an empirical characteristic-length contribution for each atom, motivated by the expectation that volume-related quantities scale with a linear dimension rather than mass directly. The scaling factor C_{LV} converts this composition-derived descriptor into an effective reference volume that is deliberately *larger* than the observed crystal volume, so that the non-negative bond-overlap corrections can be subtracted to recover the experimental volume.

The value $C_{\mathrm{LV}} = 6.1$ was determined empirically through a sensitivity analysis across multiple independent crystal-density datasets, all of which exhibited broad stable optimum regions centred near $C_{\mathrm{LV}} \approx 6$. Prediction accuracy is relatively insensitive to small variations within this region.

2.3 Bond-Overlap Dictionaries

The core of the predictor is a set of dictionaries that map bond types to effective volume-overlap values (in Å^3). Each dictionary entry corresponds to a tuple `(atom1, atom2, bond_order)`, with atom symbols alphabetically sorted. For example, a carbon–oxygen single bond is stored under the key `('C', 'O', 1)`. The overlap value represents the effective volume reduction (in Å^3) associated with that bonded arrangement, capturing systematic packing-overlap contributions.

These overlap values are derived by fitting to experimental crystal density data using Non-Negative Least Squares (NNLS):

$$\min_{\mathbf{x} \geq 0} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2$$

where $\mathbf{A}$ is the bond-count matrix, $\mathbf{y}$ is the vector of volume residuals ($V_L - V_{\mathrm{obs}}$), and $\mathbf{x}$ contains the bond-overlap coefficients. The non-negativity constraint ensures that all corrections represent volume reductions, consistent with the physical interpretation of packing-overlap contributions.

Bond descriptors are enumerated from RDKit molecular graphs with explicit hydrogen atoms included, capturing hydrogen-bonding environments. Bond descriptors occurring in fewer than ten training molecules are excluded from dictionary optimization to reduce sensitivity to poorly sampled chemical environments.

During prediction, bond environments not encountered during dictionary construction are assigned a fixed default overlap value of 0.3 Å^3, providing a conservative fallback for uncommon connectivity patterns.

Once optimized, the dictionaries are fixed. No further training or optimization occurs at prediction time.

2.4 Molecular Weight Stratification

Because bond-overlap effects scale with molecular size, a single dictionary produces systematic errors across the molecular weight spectrum. HófvarpnirHCON therefore uses up to three separate dictionaries for pure molecules, stratified by molecular weight:

- Small molecules: MW $<$ 180 g/mol
- Medium molecules: $180 \leq$ MW $<$ 400 g/mol
- Large molecules: MW $\geq$ 400 g/mol

The training procedure automatically determines the optimal stratification based on the molecular weight distribution in the training set, with a minimum class size of 150 molecules to ensure statistically robust dictionary learning. For each input molecule, the molecular weight is computed from the SMILES string (with explicit hydrogens), and the appropriate dictionary is selected automatically.

2.5 Co-crystal Prediction

For co-crystals, the input SMILES string contains a dot (e.g., `CCO.O=C(O)C`). HófvarpnirHCON splits the string on the dot and processes each component independently using the pure-molecule pipeline. The overall density is then computed as the mass-weighted average:

$$\rho_{\text{cocrystal}} = \frac{\sum_j m_j \, \rho_j}{\sum_j m_j}$$

where m_j and ρ_j are the molecular weight and predicted density of component j, respectively. No specialized co-crystal dictionaries or parameters are required; the mass-weighted average emerges naturally from volume additivity.

Note that the primary validation of HófvarpnirHCON focuses on single-component organic molecular crystals. Co-crystal prediction via mass-weighted averaging is provided as a practical extension; dedicated co-crystal dictionaries trained on multi-component data may improve accuracy for such systems.

2.6 Prediction Algorithm

The complete prediction procedure for a pure molecule is as follows:

1. Parse the SMILES string using RDKit and add explicit hydrogens.
2. Compute the molecular weight M as the sum of atomic masses.
3. Calculate the Leonardus Volume $V_L = 6.1 \sum_i m_i^{1/3}$.
4. Select the appropriate bond-overlap dictionary based on M.
5. Iterate over all bonds in the molecule:
   - Determine the atom types and bond order (1, 2, or 3).
   - Look up the overlap value δ in the dictionary (default: 0.3 Å^3 for unseen bond types).
   - Accumulate: $\Delta V_{\text{bond}} = \sum \delta_{\text{bond}}$.
6. Compute corrected volume $V_{\text{pred}} = V_L - \Delta V_{\text{bond}}$.
7. If $V_{\text{pred}} \leq 0$, set $V_{\text{pred}} = 0.5 \times V_L$.
8. Return density $\rho = M/V_{\text{pred}}$ (in g/cm^3).

For co-crystals, the algorithm recurses on each component and returns the mass-weighted density.

2.7 Implementation Details

HófvarpnirHCON is implemented in Python 3.8+ and depends only on NumPy (for numerical operations) and RDKit (for SMILES parsing and molecular graph traversal). No machine learning libraries are required for prediction. Dictionaries are stored as Python pickle files and loaded lazily on first use.

3 Validation Summary

The framework was evaluated across six published crystal-density datasets: Taniguchi (170,253 molecules), He (35,349), Davis (16,381), Jin (12,072), Taylor (1,024), and Mathieu (308). Key results include:

- Pooled molecule-weighted MAE of 0.035 g cm^{-3} across the combined benchmark population.
- Davis dataset: MAE = 0.0305 g cm^{-3}, $R^2 = 0.9457$.
- Taniguchi dataset: MAE = 0.0359 g cm^{-3}, $R^2 = 0.9213$.
- 10-fold cross-validation confirms out-of-sample generalisation across all datasets.
- Convergence analysis demonstrates that the bond-overlap dictionary stabilises within a few thousand training molecules.
- Independent half-dataset transfer experiments confirm that learned corrections represent transferable chemical relationships.

Data and Code Availability

The source code is available under the BSD-3 license. Experimental crystal density data are available from the published datasets cited in the accompanying paper: Taniguchi (2025), He (2026), Davis (2024), Jin (2023), Taylor (2024), and Mathieu (2017).

Conflict of Interest

The author declares no competing financial interest.

Contact

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